

Road Ready: A Modern Driving Education Platform 🚗

1. Project Overview

Road Ready is a comprehensive software ecosystem designed to modernize and digitize the driving education experience. It serves as a centralized hub connecting **Student Drivers** with **Professional Instructors**, replacing traditional paper logs and fragmented communication with a seamless digital solution.

The platform consists of: 1. **Web Portal:** A management system for instructors and an administrative interface for business operations. 2. **Mobile Application (React Native):** An on-the-go tool for students to track progress and for instructors to log real-time data during driving sessions.

2. Core Purpose

The primary goal of Road Ready is to improve road safety and educational efficiency. It achieves this by: * **Empowering Students:** Providing data-driven insights into their driving performance and clear paths to licensure. * **Streamlining Instructor Workflows:** Automating scheduling, booking, and administrative tasks like payout tracking. * **Modernizing Evaluation:** Using mobile device sensors (accelerometer, gyroscope, GPS) to provide objective feedback on driving skills.

3. Key Features

3.1. Diagnostic Ride Logging (Flagship Feature)

A unique system that allows students to perform a “Diagnostic Ride” using their smartphone sensors. * **Automated Scoring:** The app analyzes braking smoothness, speed compliance, and cornering stability. * **Financial Incentive:** Students who demonstrate high proficiency can skip the “Basics” module, potentially saving over **\$800** in training costs. * **Instructor Override:** For supervised rides, instructors can review automated scores and add qualitative feedback.

3.2. Integrated Student Progress Tracking

A unified dashboard available on both web and mobile that aggregates data from: * **Driving Sessions:** Detailed logs of every lesson, including road conditions, weather, and specific faults. * **Skill Analytics:** Visualization of strengths and weaknesses (e.g., “Observation,” “Space Margins”) through interactive charts. * **Milestones:** Real-time updates on practice hours and module completions.

3.3. Automated Scheduling & Availability

- **Working Hours Management:** Instructors can define recurring weekly availability.
- **Conflict Prevention:** The booking system automatically validates requested times against instructor schedules and existing appointments.
- **Real-time Booking:** Students can search for available instructors by city and budget, booking lessons instantly.

3.4. Educational Ecosystem

- **Interactive Handbooks:** Digital access to official driving guides (e.g., ICBC’s “Learn to Drive Smart”).
- **Practice Quizzes:** A database of multiple-choice questions modeled after the official 50-question knowledge test, featuring randomized sets and detailed explanations.

3.5. Financial & Communication Tools

- **Payment Integration:** Secure transaction handling via Stripe.
 - **Messaging System:** Built-in chat for direct coordination between students and instructors.
 - **Payout Management:** Automated tracking of earnings and platform fees for instructors.
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4. Technical Architecture

4.1. Backend

- **Framework:** FastAPI (Python) for high-performance, asynchronous API endpoints.
- **Database:** SQLite (SQLAlchemy ORM) for robust data persistence.
- **Security:** JWT-based authentication ensuring secure access to personal data and documents.

4.2. Frontend (Web)

- **Templating:** Jinja2 with Bootstrap 5 for a responsive, modern user interface.

- **Interactive Components:** Chart.js for progress visualization and Leaflet.js for route mapping.

4.3. Mobile

- **Framework:** React Native (Expo) for a cross-platform (iOS/Android) experience.
 - **Sensor Integration:** expo-sensors for high-frequency data collection during diagnostic rides.
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5. Summary of Implementation Status

The project has successfully moved from initial concept to a functional prototype. Recent milestones include the successful integration of the **Diagnostic Ride** backend with the mobile sensor suite and the launch of the **Consolidated Progress API**, which provides a single source of truth for student performance across all platforms.

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